

Product Installation & Setup Guide

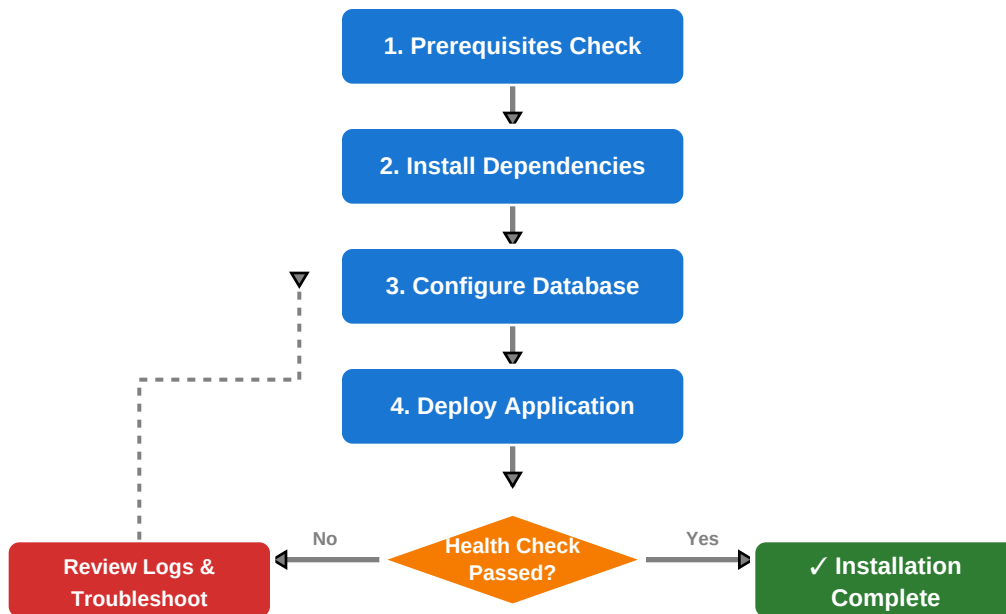
Enterprise Resolution Intelligence System v3.5

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Overview

This guide provides comprehensive instructions for installing and configuring the Enterprise Resolution Intelligence System. The installation process covers environment setup, dependency management, configuration, and initial deployment validation.

Installation Process Flow



1. Environment Prerequisites

1.1 System Requirements

Component	Minimum	Recommended	Notes
CPU	4 cores	8+ cores	x86_64 architecture
RAM	8 GB	16 GB	32 GB for production
Storage	50 GB	200 GB SSD	Separate data volume
Network	100 Mbps	1 Gbps	Low latency required

1.2 Software Dependencies

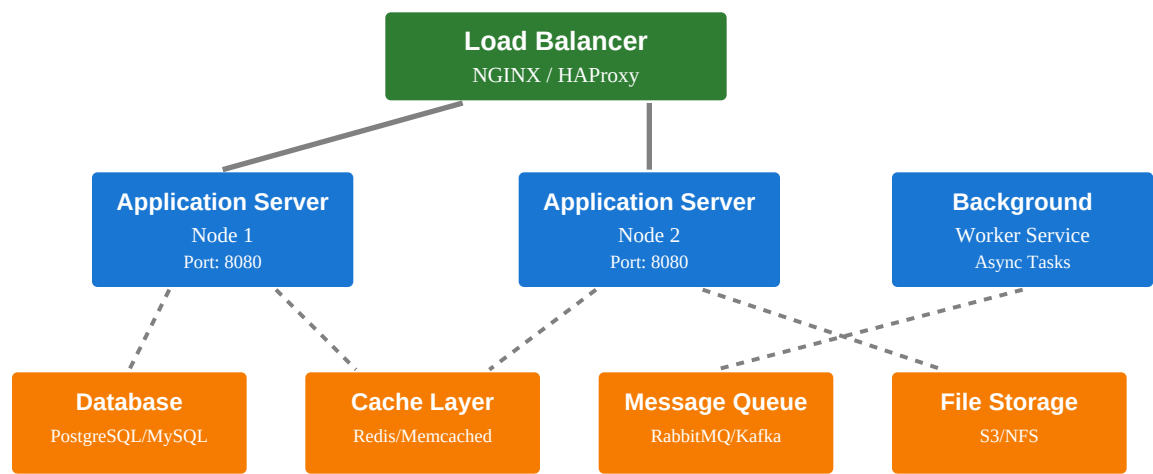
Software	Version	Purpose
Operating System	Ubuntu 22.04 LTS CentOS 8+ RHEL 8+	Host operating system
Docker	20.10+	Container runtime
Docker Compose	2.0+	Multi-container orchestration
PostgreSQL	14.x or 15.x	Primary database
Redis	7.0+	Caching and session management
NGINX	1.22+	Reverse proxy and load balancer
Node.js	18.x LTS	Runtime for web services
Python	3.10+	Backend services and ML models

1.3 Network Requirements

- Ports 80 (HTTP) and 443 (HTTPS) must be open for web access
- Port 5432 (PostgreSQL) for database connections
- Port 6379 (Redis) for cache connections
- Port 8080 (Application) for backend API
- Outbound internet access for license validation and updates

2. Deployment Steps

2.1 Deployment Architecture



2.2 Step-by-Step Installation

Step 1: System Preparation

Update system packages and install prerequisites:

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y curl wget git build-essential
```

Step 2: Install Docker

```
curl -fsSL https://get.docker.com -o get-docker.sh
sudo sh get-docker.sh
sudo usermod -aG docker $USER
```

Step 3: Install Docker Compose

```
sudo curl -L "https://github.com/docker/compose/releases/download/v2.20.0/docker-compose-$(uname -s)-$(uname -m)" -o /usr/local/bin/docker-compose
sudo chmod +x /usr/local/bin/docker-compose
```

Step 4: Download Application Package

```
wget https://releases.example.com/eris-v3.5.tar.gz
tar -xzf eris-v3.5.tar.gz
cd eris-v3.5
```

Step 5: Configure Environment Variables

```
cp .env.example .env
nano .env # Edit configuration
```

Step 6: Initialize Database

```
./scripts/init-database.sh
./scripts/run-migrations.sh
```

Step 7: Start Services

```
docker-compose up -d
```

Step 8: Verify Installation

```
docker-compose ps # Check all services are running
```

```
curl http://localhost:8080/health # Health check
```

3. Configuration Setup

3.1 Core Configuration Parameters

Parameter	Description	Default	Required
DB_HOST	Database hostname	localhost	Yes
DB_PORT	Database port	5432	Yes
DB_NAME	Database name	eris_prod	Yes
DB_USER	Database username	postgres	Yes
DB_PASSWORD	Database password	—	Yes
REDIS_HOST	Redis hostname	localhost	Yes
REDIS_PORT	Redis port	6379	Yes
APP_PORT	Application port	8080	No
LOG_LEVEL	Logging verbosity	info	No
SESSION_SECRET	Session encryption key	—	Yes
LICENSE_KEY	Product license key	—	Yes

3.2 Security Configuration

- Generate strong session secret: `openssl rand -base64 32`
- Enable SSL/TLS for production deployments
- Configure firewall rules to restrict database access
- Set up authentication for Redis if exposed

4. Common Setup Errors & Solutions

Error	Cause	Solution
Port already in use	Another service using port 8080	Stop conflicting service or change APP_PORT in .env
Database connection refused	PostgreSQL not running or incorrect credentials	Check DB_* parameters and ensure PostgreSQL is running
Permission denied	Insufficient file system permissions	Run: sudo chown -R \$USER:\$USER /opt/eris
Docker daemon not running	Docker service not started	Run: sudo systemctl start docker
Invalid license key	Expired or malformed license	Contact support for license renewal
Out of memory	Insufficient RAM allocation	Increase system RAM or adjust Docker memory limits
Migration failed	Database schema mismatch	Restore database backup and re-run migrations

5. Version Compatibility Matrix

ERIS Version	PostgreSQL	Redis	Node.js	Python
3.5.x	14.x, 15.x	7.0+	18.x	3.10, 3.11
3.4.x	13.x, 14.x	6.2+	16.x, 18.x	3.9, 3.10
3.3.x	12.x, 13.x	6.0+	16.x	3.8, 3.9
3.2.x	12.x	5.0+	14.x, 16.x	3.8

5.1 Operating System Compatibility

Operating System	Versions	Support Status
Ubuntu	20.04 LTS, 22.04 LTS	✓ Fully Supported
CentOS	8.x, 9.x	✓ Fully Supported
Red Hat Enterprise Linux	8.x, 9.x	✓ Fully Supported
Debian	11, 12	✓ Supported
Amazon Linux	2023	■ Community Support
Windows Server	2019, 2022	■ Limited (Docker Desktop)

6. Post-Installation Verification

6.1 Health Check Commands

```
# Check all services status
docker-compose ps

# Verify API health
curl http://localhost:8080/api/v1/health

# Check database connectivity
docker exec eris-db psql -U postgres -c 'SELECT version();'

# View application logs
docker-compose logs -f app
```

6.2 Access Web Interface

Once installation is complete, access the web interface at:

- HTTP: <http://<server-ip>:8080>
- HTTPS: <https://<server-ip>> (if SSL configured)

Default administrator credentials:

- Username: admin
- Password: changeme (must change on first login)

Support Contact

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